The empirical recurrence for OEIS A283570: recovering a conjecture that is not written down

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Abstract

OEIS A283570 records "Empirical recurrence of order 54", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 120$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 54, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 54 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283570 is "Number of nX6 0..1 arrays with no 1 equal to more than one of its horizontal, diagonal and antidiagonal neighbors, with the exception of exactly one element.". It has offset 1 and begins

$$12, 908, 21186, 509952, 10919674, 226897932, 4558585174, 89724600000, \ \ldots$$

The entry states:

> Empirical recurrence of order 54 (see link above)

As of the "Last modified" line on the live entry (Mar 11 2017 08:15:56, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2038$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 120$ of the 2038, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 120$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 240$ exact terms of the model returns order 54 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{54} c_i \, a(n-i),$$

c_1	30	c_{15}	-21253677096	c_{29}	-52642348555978	c_{43}	-225745183247916
c_2	-109	c_{16}	98129503223	c_{30}	-77472831803359	c_{44}	112233054977820
c_3	-2578	c_{17}	230415959836	c_{31}	30799302506252	c_{45}	154111335406946
c_4	10304	c_{18}	71989927271	c_{32}	118676237784755	c_{46}	-113858099340601
c_5	51682	c_{19}	-754417800718	c_{33}	26454102730446	c_{47}	-56922082650866
c_6	-549441	c_{20}	-2074694429286	c_{34}	-169654620092884	c_{48}	60284283599464
c_7	-655698	c_{21}	620728686340	c_{35}	-94198039213980	c_{49}	9645486116894
c_8	12265369	c_{22}	5660226306173	c_{36}	227259046559949	c_{50}	-16850539628576
c_9	-798008	c_{23}	4563869873370	c_{37}	103536750049866	c_{51}	97890008312
c_{10}	-189652210	c_{24}	-5728620921456	c_{38}	-205520999825307	c_{52}	2065662929039
c_{11}	161955652	c_{25}	-17296599375552	c_{39}	-106333584441504	c_{53}	-83149693176
c_{12}	2648743654	c_{26}	-9255340269664	c_{40}	99796325117302	c_{54}	-95177186064
c_{13}	-660369902	c_{27}	38804241323502	c_{41}	187947582618524		
c_{14}	-24275319158	c_{28}	40711802879386	c_{42}	-69766146555137		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 54, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $54 + 4$ terms removed returns order 54 as well, so $\deg q_{\min} = 54 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $54+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 54 that this sequence satisfies — and that family turns out to have exactly one member.

References

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